WORKED SOLUTION

STEP III 2001, Question 2

Pure • Hyperbolic Substitution and Area

STEP III 2001, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q2

The key idea

Both inequalities say "stay on one side of a rectangular-type hyperbola", so the first job is to sketch the two hyperbolas and find where they cross. The region turns out to be a bounded, four-fold symmetric patch, and its area is a pair of integrals of the form $\int \sqrt{x^2 \pm a^2}\,\mathrm{d}x$. Those are exactly the integrals that hyperbolic substitutions tame, and the first part is there so that we can turn the resulting $\cosh^{-1}$ (and $\sinh^{-1}$) values into logarithms.

Solution

The logarithmic form of $\cosh^{-1} x$

Let $x \geq 1$ and $y = \cosh^{-1} x$, which by definition is the *non-negative* number with $\cosh y = x$. Then

$$x = \frac{\mathrm{e}^y + \mathrm{e}^{-y}}{2} \quad \Longrightarrow \quad \mathrm{e}^{2y} - 2x\,\mathrm{e}^y + 1 = 0,$$

a quadratic in e^y, so

$$\mathrm{e}^y = x \pm \sqrt{x^2 - 1}.$$

Ask yourself Which sign do we take, and why?

The two roots multiply to 1 (the constant term of the quadratic), and the root with the plus sign is at least 1. So the root with the minus sign is at most 1, which would give $y \leq 0$. Since $y \geq 0$ we must take the plus sign (when $x = 1$ the two roots coincide anyway), and so

$$\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1}),$$

as required. Exactly the same argument with $x = \sinh y$ gives $\mathrm{e}^{2y} - 2x\,\mathrm{e}^y - 1 = 0$; now the roots multiply to -1, so only the plus root is positive, and $\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$. We shall use both.

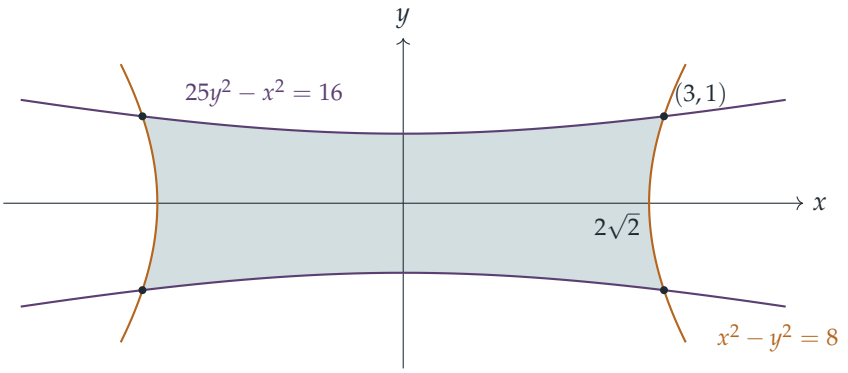
The region

Rewrite the inequalities as

$$x^2 - y^2 \le 8 \qquad \text{and} \qquad 25y^2 - x^2 \le 16.$$

The curve $x^2 - y^2 = 8$ is a hyperbola with branches opening left and right through $(\pm 2\sqrt{2}, 0)$; the first inequality keeps us *between* its branches, i.e. $|x| \le \sqrt{y^2 + 8}$. The curve $25y^2 - x^2 = 16$ is a hyperbola with branches opening up and down through $(0, \pm\frac{4}{5})$; the second inequality keeps us between those branches, i.e. $|y| \le \frac{1}{5}\sqrt{x^2 + 16}$.

The curves cross where $x^2 = y^2 + 8$ and $25y^2 = x^2 + 16 = y^2 + 24$, so $y^2 = 1$ and $x^2 = 9$: at the four points $(\pm 3, \pm 1)$.



The region (shaded) is bounded: if (x, y) is in it then $x^2 \le y^2 + 8 \le \frac{1}{25}(x^2 + 16) + 8$, so $x^2 \le 9$, and then $y^2 \le \frac{25}{25} = 1$. Both inequalities are unchanged by $x \mapsto -x$ and $y \mapsto -y$, so the region is symmetric in both axes and its area is four times the area A of the part in the first quadrant.

In the first quadrant we slice vertically.

- For $0 \le x \le 2\sqrt{2}$ we have $x^2 - 8 \le 0 \le y^2$, so the first inequality holds automatically, and y runs from 0 up to $\frac{1}{5}\sqrt{x^2 + 16}$.
- For $2\sqrt{2} \le x \le 3$, y runs from $\sqrt{x^2 - 8}$ up to $\frac{1}{5}\sqrt{x^2 + 16}$ (and the lower curve is below the upper one exactly when $x^2 \le 9$, by the calculation of the crossing points).

Hence

$$A = \frac{1}{5}\int_0^3 \sqrt{x^2 + 16}\,\mathrm{d}x \; - \; \int_{2\sqrt{2}}^3 \sqrt{x^2 - 8}\,\mathrm{d}x = \tfrac{1}{5}I_1 - I_2.$$

The first integral

Substitute $x = 4\sinh s$, so $\mathrm{d}x = 4\cosh s\,\mathrm{d}s$ and $\sqrt{x^2 + 16} = 4\cosh s$ (positive, since $\cosh s > 0$). The limits are $s = 0$ and $s = S = \sinh^{-1}\frac{3}{4} = \ln\left(\frac{3}{4} + \frac{5}{4}\right) = \ln 2$. Using

$\cosh^2 s = \frac{1}{2}(\cosh 2s + 1)$,

$$I_1 = \int_0^S 16\cosh^2 s\,\mathrm{d}s = \int_0^S 8(\cosh 2s + 1)\,\mathrm{d}s = \Big[8\sinh s\cosh s + 8s\Big]_0^S$$
$$= 8\cdot\tfrac{3}{4}\cdot\tfrac{5}{4} + 8\ln 2 = \tfrac{15}{2} + 8\ln 2,$$

since $\sinh S = \frac{3}{4}$ and $\cosh S = \sqrt{1+\frac{9}{16}} = \frac{5}{4}$.

The second integral

Substitute $x = 2\sqrt{2}\cosh t$ with $t \geq 0$, so $\mathrm{d}x = 2\sqrt{2}\sinh t\,\mathrm{d}t$ and $\sqrt{x^2-8} = 2\sqrt{2}\sinh t$ (positive, since $t \geq 0$). The limits are $t = 0$ and

$$t = T = \cosh^{-1}\frac{3}{2\sqrt{2}} = \ln\Big(\frac{3}{2\sqrt{2}} + \sqrt{\tfrac{9}{8}-1}\Big) = \ln\frac{4}{2\sqrt{2}} = \ln\sqrt{2} = \tfrac{1}{2}\ln 2,$$

by the first part. Using $\sinh^2 t = \frac{1}{2}(\cosh 2t - 1)$,

$$I_2 = \int_0^T 8\sinh^2 t\,\mathrm{d}t = \int_0^T 4(\cosh 2t - 1)\,\mathrm{d}t = \Big[4\sinh t\cosh t - 4t\Big]_0^T$$
$$= 4\cdot\frac{1}{2\sqrt{2}}\cdot\frac{3}{2\sqrt{2}} - 2\ln 2 = \tfrac{3}{2} - 2\ln 2,$$

since $\cosh T = \frac{3}{2\sqrt{2}}$ and $\sinh T = \sqrt{\frac{9}{8}-1} = \frac{1}{2\sqrt{2}}$.

Putting it together

$$A = \tfrac{1}{5}\Big(\tfrac{15}{2} + 8\ln 2\Big) - \Big(\tfrac{3}{2} - 2\ln 2\Big) = \tfrac{3}{2} + \tfrac{8}{5}\ln 2 - \tfrac{3}{2} + 2\ln 2 = \tfrac{18}{5}\ln 2.$$

The algebraic parts cancel exactly, and the total area is

$$4A = \frac{72}{5}\ln 2,$$

as required.

Checks

- $\frac{72}{5}\ln 2 \approx 9.981$. Simpson's rule on the two integrals gives $I_1 \approx 13.0452$ and $I_2 \approx 0.11371$, matching $\frac{15}{2} + 8\ln 2$ and $\frac{3}{2} - 2\ln 2$; and simply counting points of a fine grid that satisfy both inequalities gives an area of about 9.98.
- Sanity check on size: the region fits inside the rectangle $[-3,3]\times[-1,1]$ of area 12, and it misses the corners of that rectangle only by thin slivers, so an area just under 10 is about right.
- $\cosh^{-1} 1 = \ln 1 = 0$, as it should be.

Common slips

- Reading the inequalities as "outside" the hyperbolas: $y^2 \geq x^2 - 8$ is the region *between* the branches of $x^2 - y^2 = 8$, which contains the y-axis.
- Forgetting that the first inequality only bites for $x > 2\sqrt{2}$, and so subtracting $\int_0^3 \sqrt{x^2 - 8}\,\mathrm{d}x$, which is not even real.
- Taking the minus sign in $\mathrm{e}^y = x \pm \sqrt{x^2 - 1}$, which gives $-\cosh^{-1} x$.

Beyond STEP

The two families $\int \sqrt{x^2 + a^2}\,\mathrm{d}x$ and $\int \sqrt{x^2 - a^2}\,\mathrm{d}x$ are handled by hyperbolic substitutions for the same reason $\int \sqrt{a^2 - x^2}\,\mathrm{d}x$ is handled by $x = a\sin\theta$: the identity $\cosh^2 - \sinh^2 = 1$ removes the square root. In fact the hyperbolic functions measure *area*: the parameter t in $(\cosh t, \sinh t)$ is twice the area of the sector of the unit hyperbola, just as θ is twice the area of a circular sector. Siklos, *Advanced Problems in Mathematics*, has many more questions of this flavour, and Bartle and Sherbert, *Introduction to Real Analysis*, shows how the exponential, logarithm and hence the hyperbolic functions are put on a rigorous footing.

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